altairsim — The SD Systems SBC-200 -- a Z80 single-board computer

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The **SD Systems SBC-200** was a whole computer on one S-100 board: a 4 MHz Z80, an Intel 8251 serial console, a Z80-CTC baud generator, a parallel port, RAM, and boot-PROM sockets, acting as the system's bus master. It shipped with the **SD/MS monitor** in EPROM and, with a VersaFloppy controller, booted SDOS or CP/M.

```
cd examples/sdsys
altairsim sbc200.toml
(press Enter)   ->   .
```

This is a thin delta on the built-in `sbc200` machine (`altairsim sbc200`): the 4 MHz Z80, the 8251 console at ports 7C/7D, the **MSMONR21** monitor in EPROM at E000, the SD **DDBIOS** disk BIOS at F000, and RAM filling the rest of the 64K.

Press Enter to get the prompt

The monitor prints nothing until you press Enter, and that is authentic. The SBC-200 wires the 8251's receive-data line to its /DSR input, and MSMONR21 uses that to **auto-detect your baud rate**: on reset it waits, then times the start bit of the first character you type (a carriage return), reading status bit 7 in a tight loop. Press Enter and it matches your speed and prints its `.` prompt.

The simulator models that line in emulated processor time, so the genuine ROM auto-bauds on the console with no changes. (If a run ever seems to "hang" at startup, it is waiting for that first CR -- press Enter.)

The monitor

At the `.` prompt, MSMONR21 gives you a full Z80 monitor -- memory display/edit, fill, move, search, I/O port access, breakpoints and single-step, and hex arithmetic. A few to try:

```
.D E000 E01F        display the monitor ROM, hex + ASCII
.E 8000             examine/substitute memory at 8000
.H 1234 0100        hex arithmetic: sum then difference
```

Type `.` to abort a command back to the prompt. The full command set is in `reference/SD Systems Monitor.md`.

Booting SDOS

`sdos.toml` adds a **VersaFloppy II** floppy controller with a bootable SDOS master disk in drive A. From the monitor prompt, `C` cold-boots the OS:

```
cd examples/sdsys
altairsim sdos.toml
(press Enter)   ->   .
C               ->   cold-boot SDOS from drive A
```

```
32K SD-OS Version 1.8B
DELTEC ENTERPRISES LLC

[A]
```

`[A]` is the SDOS prompt. The monitor's other disk commands work too: `R`/`W` read and write 128/256-byte sectors, `Z` formats a diskette. The disk is `SDOS-18B-SSDDR-256-32K-MASTER.DSK` -- an 8" single-sided, double-density, 256-byte-sector image (26 sectors x 77 tracks), sysgen'd for a 32K system. It is mounted **read/write**, so SDOS can save files; run `git checkout` on it to restore the master if you change it.

What is not here yet

The Z80-CTC baud generator, the parallel port, and the SBC-200 memory switch-out are later phases. The serial console and the VersaFloppy disk both work today.